

PAPER-1

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **26** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

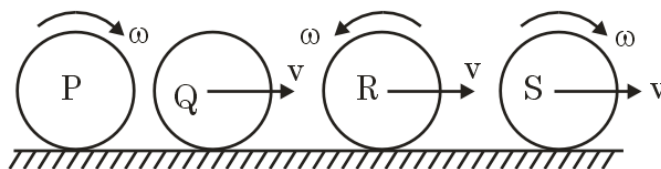
Please see the last page of this booklet for rest of the instructions

PHYSICS

SEC-1 : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : -1 In all other cases

1. Four solid spheres are made to move on a rough horizontal surface. Sphere P is given a spin and released. Sphere Q is given a forward linear velocity. Spheres R and S are given linear and rotational motions as shown in the figure. Directions of the friction force on spheres P, Q, R, S are respectively



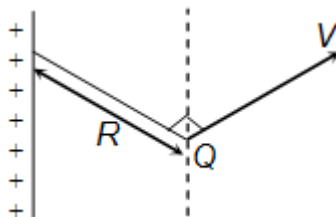
- (A) Right, Left, Right, Left (B) Right, Left, Left, Right
 (C) Left, Right, Left, either Left or Right (D) Right, Left, Left, either Left or Right
2. A non-conducting solid sphere has a non-concentric spherical cavity in it. Remaining sphere has uniform volume charge density. A proton is given some velocity in the cavity. What can be its possible path of trajectory?
- (A) Only straight line (B) Only parabola
 (C) Either straight line or parabola (D) Either straight line or circle.

Space for Rough Work

3. A light source is placed at a distance b from a screen. The power of the lens required to obtain k -fold magnified image is

(A) $\frac{k+1}{kb}$ (B) $\frac{(k+1)^2}{kb}$ (C) $\frac{kb}{k+1}$ (D) $\frac{kb}{(k-1)^2}$

4. A point charge $+Q$ is projected with velocity V from a distance R from an infinitely long fixed line charge of linear charge density λ . What is the angular velocity of the point charge when its radial distance from the line charge becomes eR ? (direction V of is perpendicular to the wire and e is positive constant)



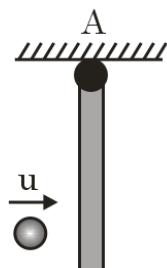
(A) $\frac{V}{eR}$ (B) $\frac{eV}{R}$ (C) $\frac{e^2V}{R}$ (D) $\frac{V}{e^2R}$

Space for Rough Work

SEC-2 : (Maximum Marks: 32)

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Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

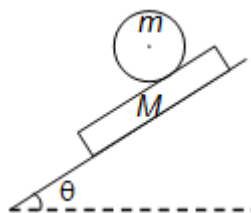
1. In the given figure a ball strikes a rod elastically and rod is hinged smoothly at point A. Then which of the statement(s) is/are correct for the collision?



- (A) Linear momentum of system (ball + rod) may be conserved.
 (B) Angular momentum of system about hinged point A is conserved.
 (C) Kinetic energy of system is same just before and just after collision.
 (D) Linear momentum of ball may be conserved.

Space for Rough Work

2. A solid sphere of mass m is released on the plank of mass M which lies on an inclined plane of inclination θ as shown in figure. There is sufficient frictional force between sphere and plank and the minimum value of co-efficient of friction between plank and surfaces is μ to keep the plank at rest. Then



- (A) frictional force between sphere and plank is $\frac{2}{7}mg \sin \theta$, when plank is at rest.
- (B) the value of μ is $\frac{7M + 2m}{7(M + m)} \tan \theta$.
- (C) If there is no friction between the plank and inclined plane, then acceleration of plank is less than $g \sin \theta$.
- (D) If there is no friction between plank and inclined plane, then friction force on the sphere is zero.
3. A charged particle q is placed at a distance d from the center of neutral conducting sphere of radius $R(<d)$, then in static condition at the center of sphere
- (A) magnitude of electric field due to induced charge is $\frac{kq}{d^2}$
- (B) magnitude of electric potential due to induced charge is $-\frac{kq}{R}$
- (C) magnitude of electric field due to induced charge is zero
- (D) magnitude of electric field due to charge q is $\frac{kq}{d^2}$ (where $k = \frac{1}{4\pi\epsilon_0}$)

Space for Rough Work

4. A particle of mass m is moving in a field where the potential energy is given by $U(x) = U_0(1 - \cos ax)$, where U_0 and a are constants and x is the displacement from mean position. Then (for small oscillations)

(A) The time period is $T = 2\pi\sqrt{\frac{m}{aU_0}}$.

(B) The speed of particle is maximum at $x = 0$.

(C) The amplitude of oscillations is $\frac{\pi}{a}$.

(D) The time period is $T = 2\pi\sqrt{\frac{m}{a^2U_0}}$

5. The concave and convex surfaces of a thin concave-convex lens of refractive index 1.5 has radius of curvature 50 and 10 cm respectively. The concave side is silvered and placed on a horizontal surface as shown



- (A) focal length of lens is 25 cm
 (B) focal length of the combination (silvering) is 25 cm
 (C) image of the object at 50 cm will coincide with itself
 (D) image will not coincide in this type of silvered lenses

Space for Rough Work

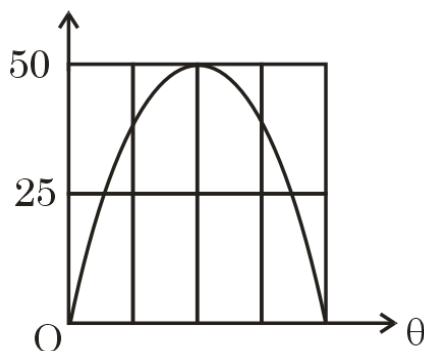
6. A thin equiconvex spherical glass lens ($\mu = 3/2$) of radius of curvature 30 cm is placed on the x -axis with its optical centre at $x = 40$ cm and principal axis coinciding with the x -axis. A light ray given by the equation $39y = -x + 1$ (x and y are in cm) is incident on the lens, in the direction of positive x -axis. Then choose the correct alternative(s).
- (A) The equation of refracted ray is $39y = x + 1$
- (B) The equation of refracted ray is $130y = x - 170$
- (C) The equation of refracted ray if space on right side of the lens is filled with a liquid of refractive index $4/3$ is $390y + x + 350 = 0$.
- (D) The equation of refracted ray if space on right side of the lens is filled with a liquid of refractive index $4/3$ is $390y - x + 350 = 0$.
7. Three planets of same density have radii R_1 , R_2 and R_3 such that $R_1 = 2R_2 = 3R_3$. The gravitational field at their respective surfaces are g_1 , g_2 and g_3 and escape velocities from their surfaces are v_1 , v_2 and v_3 . then
- (A) $\frac{g_1}{g_2} = 2$ (B) $\frac{g_1}{g_3} = 3$ (C) $\frac{v_1}{v_2} = \frac{1}{4}$ (D) $\frac{v_1}{v_3} = 3$
8. A particle of mass m moves in a straight line. If v is the velocity at a distance x from a fixed point on the line and $v^2 = a - bx^2$ where a and b are constant, then
- (A) the motion continues along the positive x -direction only
- (B) the motion is simple harmonic
- (C) the particle oscillates with a frequency equal to $\frac{\sqrt{b}}{2\pi}$
- (D) the total energy of the particle is ma

Space for Rough Work

SEC-3 : (Maximum Marks: 18)

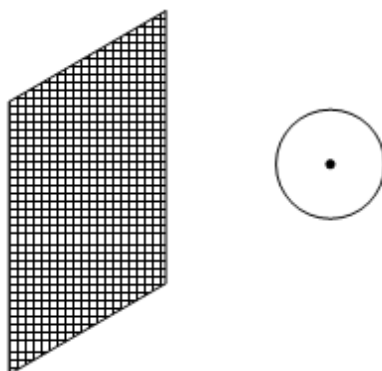
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- Answer to each question will be evaluated according to the following marking scheme:
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- The gravitational potential energy of a satellite revolving around the earth in circular orbit is -4 MJ. Find the additional energy (in MJ) that should be given to the satellite so that it escapes from the gravitational field of earth. Assume earth's gravitational force to be the only gravitational force on the satellite and no atmospheric resistance.
- An electric dipole is placed in a uniform electric field $\vec{E}$ of magnitude 40 N/C. Graph shows the magnitude of the torque (10^{-28} Nm) on the dipole versus the angle θ between the field $\vec{E}$ and the dipole moment $\vec{p}$. The magnitude of dipole moment $\vec{p}$ is equal to $x \times 10^{-28}$ C-m. Find x .

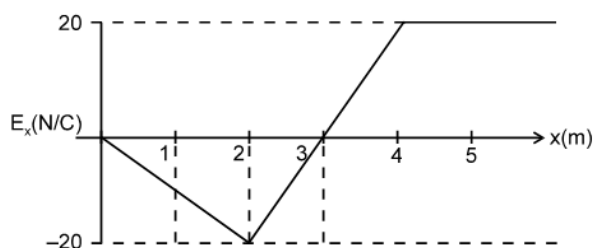


Space for Rough Work

3. Let there be a non-conducting infinite large sheet of charge of surface charge density $\frac{1}{36\pi} \text{ nC/m}^2$. An uncharged conducting spherical shell of radius 20 cm is placed in front of sheet. A point charge 6mC is placed at centre of the shell. Force on point charge due to shell is x mN. Find value of x

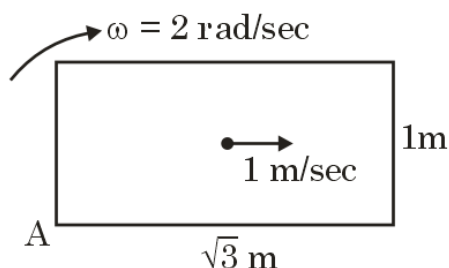


4. A graph of the x component of the electric field as a function of x in a region of space is shown. The Y and Z components of the electric field are zero in this region. If the electric potential is 10 V at the origin, then potential at x = 2.0 m is :

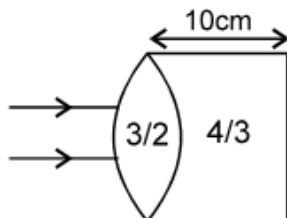


Space for Rough Work

5. A uniform rectangular plate of mass 1 kg is performing combined rotational motion and translational motion with constant angular speed 2 rad./sec (as shown in figure). Speed of centre of mass of plate is 1 m/sec. Find the kinetic energy (in J) of plate with respect to a point A (point A is fixed on the plate) :



6. A parallel paraxial beam of light is incident on the arrangement as shown. $\mu_A = 3/2$, $\mu_B = 4/3$, the two spherical surfaces are very close and each has a radius of curvature 10 cm. Find the distance (in cm) of point where the rays are focussed. (w.r.t. point of entry)



Space for Rough Work

CHEMISTRY

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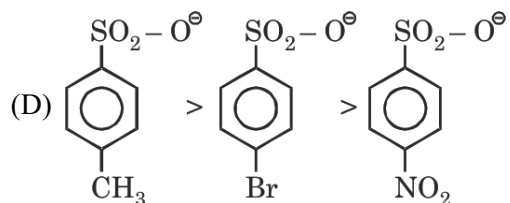
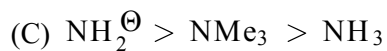
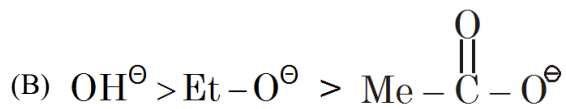
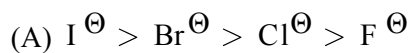
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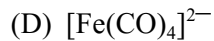
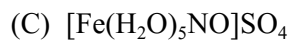
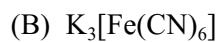
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1. 500 mL of methane (CH_4) effuses through a small hole in 48 sec. How much time will be taken by same volume of helium to effuse out of the same orifice ?
- (A) 192 sec (B) 24 sec (C) 12 sec (D) 96 sec
2. What is the osmotic pressure of decimolar solution of glucose at 30°C ?
- (A) 2.48 atm (B) 8.24 atm (C) 3.48 atm (D) 0.241 atm
-

Space for Rough Work

3. Select correct order of leaving groups



4. Which of the following complex show synergic bonding as well as follow Sidgwick rule of EAN.



Space for Rough Work

SEC-2 : (Maximum Marks: 32)

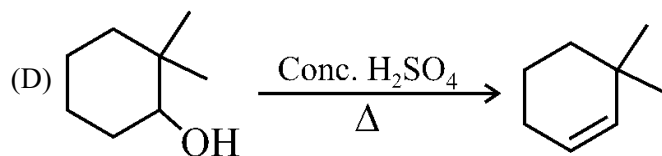
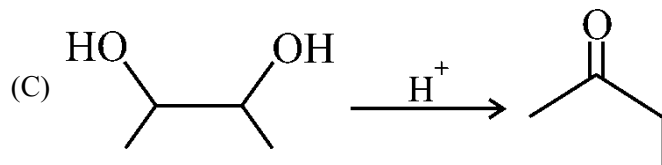
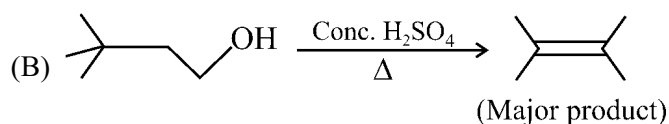
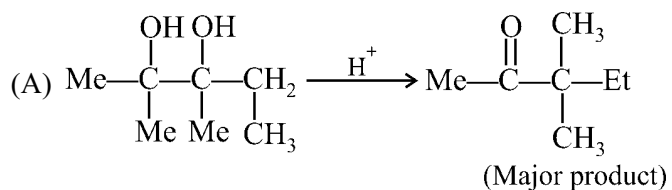
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- Potassium crystallizes in a body centred cubic lattice with edge length $a = 5.0 \text{ \AA}$. Then choose the correct statements in given option -
 (A) The number of nearest neighbour to any K atom is 8
 (B) The number of next nearest neighbour to any K atom is 6.
 (C) Distance between nearest neighbours is $4.33 \text{ \AA}$
 (D) Distance between next nearest neighbours is $5.0 \text{ \AA}$
- In the decay process correct option(s) is/are :

$$A \xrightarrow{-\alpha} B \xrightarrow{-\beta} C \xrightarrow{-\beta} D$$
 (A) A & B are isobar
 (B) A & D are isotopes
 (C) B, C & D are isobars
 (D) A & C isotones

Space for Rough Work

3. Select correct statement(s) from the following
- (A) For ideal gas in isobaric process $dU = nC_v dT$
- (B) For ideal gas in isochoric process $dU = nC_v dT$
- (C) For the reaction $A_{(l)} \rightarrow A_{(g)}$ at constant T & P, $\Delta U = 0$
- (D) Heat capacity is a state function
4. Select the reaction(s) giving correct major product :

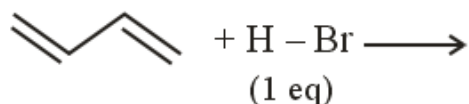


Space for Rough Work

5. select the incorrect statement(s) among following :

- (A) Me-O^- is better nucleophile than Me-S^- in H_2O
- (B) Addition of hydrohalogenation at alkene follow non classical carbocation mechanish
- (C) Benzene decolourise bromine water ($\text{Br}_2 / \text{H}_2\text{O}$) solution
- (D) DMF, DMSO are the example of polar protic solvent

6. Incorrect statement regarding product of reaction :



- (A) Kinetic controlled product is a secondary bromide whereas thermodynamically controlled product is a tertiary bromide
- (B) Kinetic controlled product is a secondary bromide whereas thermodynamically controlled product is a primary bromide
- (C) Both kinetic & thermodynamic controlled product is secondary bromide
- (D) Both kinetic & thermodynamic controlled product is a tertiary bromide

Space for Rough Work

7. A homodiatomic molecule ' X_2 ' have bond order '1', then which of the following prediction(s) will be correct for ' X_2 '
- (A) The difference of total number of bonding and antibonding e^- is 2
- (B) Molecule can be odd electron molecule
- (C) Total number of bonding electrons is more than antibonding electrons
- (D) Last e^- of ' X_2 ' molecule must be entered into bonding molecular orbital
8. Select the correct IUPAC Name from following :
- (A) Diamminesilver(I) chloride : $[Ag(NH_3)_2]Cl$
- (B) Potassium hexacyanidoferrate(III) : $K_3[Fe(CN)_6]$
- (C) Tetraamminecopper(II) sulphate : $[Cu(NH_3)_4]SO_4$
- (D) Hexaamminecobalt(III) chloride : $[Co(NH_3)_6]Cl_3$
-

Space for Rough Work

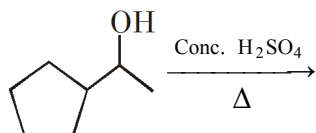
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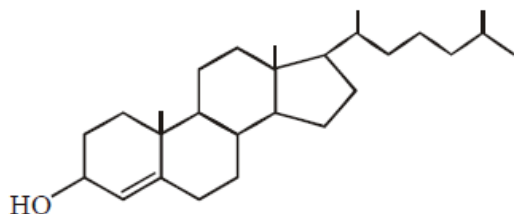
1. The reaction $2A + B \rightarrow C + D$ and follows the rate law $-\frac{d[A]}{dt} = K[A]^2[B]$. The sum of values of x and y in the following data:

Set	$[A_0] \times 10^6 \text{ M}$	$[B_0] \times 10^6 \text{ M}$	half-life (s)
1	3×10^5	40	62.5
2	3×10^5	60	x
3	5	3×10^5	625
4	10	3×10^5	y

2. Identify number of α -hydrogen present in final major product in given reaction



3. Calculate total number of chiral centre present in following compound :



Space for Rough Work

4. Numbers of elements from the following having higher IE than oxygen are.
He, Ne, N, Be, C, F
5. Find the number of ligands which behave predominantly as π -donor ligand (as well as non-classical ligand)
 C_2H_4 , C_6H_6 , NO^+ , NO, CO, NH_3 , NH_2^- , H_2O , OH^-
6. Which of the following pair having correct order of covalent character
($\text{AlCl}_3 > \text{MgCl}_2$), ($\text{BeCl}_2 > \text{LiCl}$), ($\text{PbCl}_4 > \text{PbCl}_2$), ($\text{HgS} > \text{ZnS}$), ($\text{NiCl}_2 > \text{CaCl}_2$)
-

Space for Rough Work

MATHEMATICS

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-
- Let $f(1) = a$, $f(3) = a^2$ & $f(x) \leq 6 \forall x \in [1, 3]$, then greatest possible value of 'a' is –
 (A) 2 (B) 4
 (C) 5 (D) 6
 - Number of integral value of 'k' for which the equation $4\sin^{-1}(|x|) = k$ has exactly two solutions, is/are
 (A) 3 (B) 4
 (C) 5 (D) 6
-

Space for Rough Work

3. If $\log_2 \sin x - \log_2 \cos x - \log_2 (1 - \tan^2 x) = -1$, then number of solutions of the equation for $x \in [0, 2\pi)$ are -

(A) 1 (B) 2 (C) 3 (D) 4

4. Let $f(x) = \begin{vmatrix} x & \cos x & 1 \\ 2x+1 & e^x - x + 1 & 2 \\ 3x+2 & \sec x + 2 & 3 \end{vmatrix}$, then $\lim_{x \rightarrow 0} \frac{f(x)}{x^2}$ is equal to -

(A) -1 (B) 0 (C) 1 (D) $\frac{1}{2}$

Space for Rough Work

SEC-2 : (Maximum Marks: 32)

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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

-
1. Consider function $f(x) = \begin{cases} x & , x < 2 \\ k & , x = 2 \\ x^2 - 2x + 4 & , x > 2 \end{cases}$. Which of the following statement/s is/are true ?
- (A) for $k = 0$; $x = 2$ is local minima
- (B) if $f(x)$ is strictly increasing then $k \in [2, 4]$
- (C) least integral value of k for which $x = 2$ is local maxima is 5
- (D) Range of $f(x)$ is $(-\infty, \infty)$
2. The abscissa of the point on the curve $\sqrt{xy} = a + x$, the tangent at which cuts off equal intercepts from the co-ordinate axes is ($a > 0$)
- (A) $\frac{a}{\sqrt{2}}$ (B) $-\frac{a}{\sqrt{2}}$ (C) $a\sqrt{2}$ (D) $-a\sqrt{2}$
-

Space for Rough Work

3. Let $f : (0, \infty) \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ be a function defined as $f(x) = \tan^{-1}(\ln x)$ and g be the inverse function of f , then
- (A) $g'(0) = g(0)$ (B) $f'(1) = g'(0)$
 (C) $g'(0) + g''(0) = 2$ (D) $g'(0) + g''(0) = 1$
4. The value of $\log_{10} \left((\sin 18^\circ + \cos 36^\circ) \left(\tan \frac{3\pi}{8} - \tan \frac{\pi}{8} \right) (\cot 75^\circ + \cot 15^\circ)^{\frac{1}{4}} \right)$ equals
- (A) additive inverse of $\sin(1050^\circ)$
 (B) multiplicative inverse of $\operatorname{cosec}(150^\circ)$
 (C) $\sin \frac{3\pi}{10} - \cos \frac{2\pi}{5}$
 (D) $\log_x 2$ where $x = \sqrt{12 + \sqrt{12 + \sqrt{12 + \dots \infty}}}$

Space for Rough Work

5. Which of the following is not always correct
- (A) A tangent can be drawn to the graph of the function $y = f(x)$ at $x = a \Rightarrow f(x)$ is derivable at $x = a$
- (B) If $f(x)$ is derivable at $x = a \Rightarrow f'(x)$ is continuous at $x = a$
- (C) If $f(x)$ is derivable and continuous in (a, b) then $f'(c) = \frac{f(b) - f(a)}{b - a}$ for some $c \in (a, b)$
- (D) If $f(x)$ is increasing in $(a - h, a)$ and decreasing in $(a, a + h)$ has a maximum at $x = a$
6. If $f(x) = \frac{2021^x - 2021^{-x}}{2021^x + 2021^{-x}}$ then
- (A) Range of $f(x)$ is $(-1, 1)$
- (B) $f(x)$ is even function
- (C) $f(x)$ is odd function
- (D) domain of $f(x)$ is $\mathbb{R}$

Space for Rough Work

7. Let $f(x) = \max \{\lambda - |x|, x^2 - \lambda^2\}$, then (where $\lambda \in \mathbb{R}$)
- (A) $f(x)$ is differentiable $\forall x \in \mathbb{R}$ for two integral values of ' λ '.
- (B) $f(x)$ is non differentiable in $\mathbb{R}$, if $\lambda \in [-1, 0]$
- (C) $f(x)$ attains three points of extremum, if $\lambda < -1$
- (D) $f(x)$ always has its global minimum value $\forall \lambda \in \mathbb{R}$
8. The equation $\frac{\log_8 (8/x^2)}{(\log_8 x)^2} = 3$ has
- (A) No integral solution (B) One natural solution
- (C) Two real solutions (D) One irrational solution
-

Space for Rough Work

SEC-3 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

1. If the expression $a\cos^2x + b$ ($a > 0$) lies in the interval $[-3, 8]$ for every $x \in \mathbb{R}$, then the value of $(a + 2b)$ is equal to :-
2. If the curves $y = x^3 + ax$ and $y = bx^2 + c$ pass through the point $(-1, 0)$ and have a common tangent line at this point then the value of $|\sin^{-1}(a) + \sin^{-1}(\sin b) + \cos^{-1}(c)|$ is
3. If the length of sub-tangent to the curve $x^2y^2 = 81$ at the point $P(\alpha, \beta)$ is 3 then absolute value of sum of all possible value of $(\alpha + \beta)$ is

Space for Rough Work

4. Let $x = \sin t$, $y = \sin(kt)$; where t is a parameter and k is constant such that $(1 - x^2)y_2 - \lambda_1 xy_1 + k^2 y \lambda_2 = 0$ then $\frac{\lambda_1}{2\lambda_2}$ is $(y_1 \neq 0)$
(Where y_n denotes n^{th} order derivative)
5. If the range of real-valued function $f(x) = \sqrt{\cos^{-1}(2x) + 2} + \sqrt{1 - \cos^{-1}(2x)}$ is $[a, b+1]$, then $(a + b)$ is equal to
6. If $f(x) = \frac{x}{\ln x}$ satisfies Rolle's theorem in $[a, b]$, $a, b \in \mathbb{N}$, then the value of $(a + b)$ is
-

Space for Rough Work

Space for Rough Work